D.A.A Lab Assignment 10:

AP251100010807

S.Jignesh

Q1– Implementation of 0/1 knapsack Code:

main.cpp

```cpp

#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int knapsack(int capacity, vector<int>& weights, vector<int>& values) {
    int n = weights.size();
    vector<int> dp(capacity + 1, 0);

    for (int i = 0; i < n; i++) {
        for (int w = capacity; w >= weights[i]; w--) {
            dp[w] = max(dp[w], values[i] + dp[w - weights[i]]);
        }
    }

    return dp[capacity];
}

int main() {
    int n, capacity;
    cin >> n >> capacity;

    vector<int> weights(n), values(n);

    for (int i = 0; i < n; i++) {
        cin >> weights[i];
    }

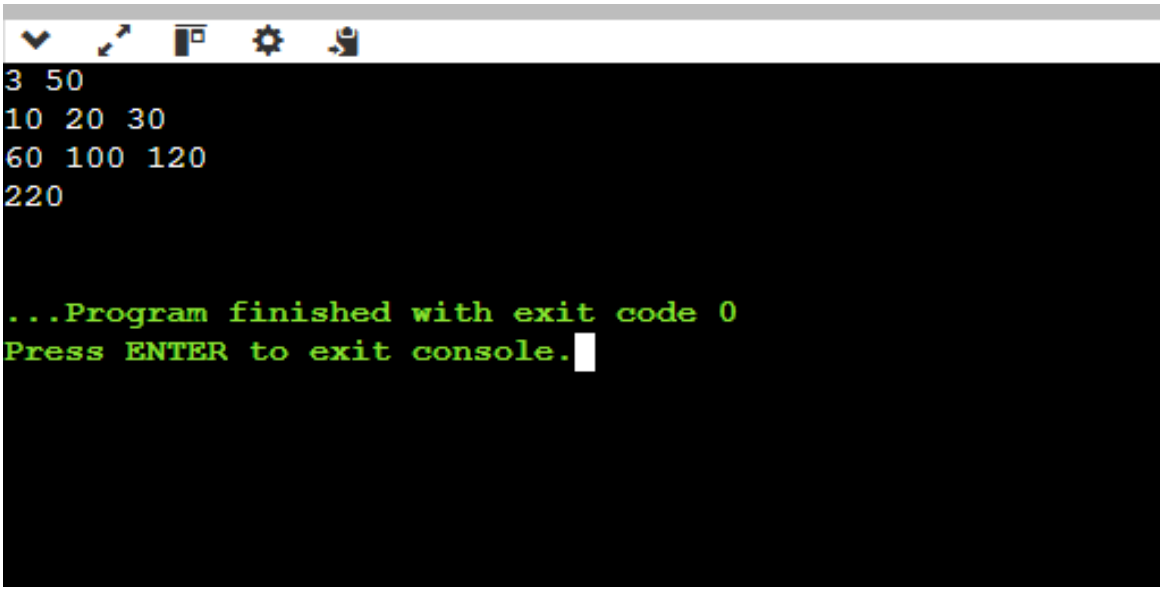
    for (int i = 0; i < n; i++) {
        cin >> values[i];
    }

    cout << knapsack(capacity, weights, values) << endl;

    return 0;
}

```

Output:


3 50
10 20 30
60 100 120
220
...Program finished with exit code 0
Press ENTER to exit console.